

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 23

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Any 3 from:</i> • Raise river banks / build sea walls, flood barriers or wave screens • Permeable urban surfaces (gravel, soil) instead of concrete / tarmac to slow runoff • River flow management: upstream dams / low soil dams / afforestation / maintained flood plains to delay runoff • Raised buildings on stilts or floating houses • Pumping rainwater from land below river level • Managed retreat - abandoning land where protection costs exceed value (eg Happisburgh)	3
01.2	• Where the cost of flood prevention is greater than the value of what is protected, or defences would be ineffective • Land is abandoned as water levels rise / coast erodes; defences used only where land is very important	2
01.3	<i>Any 2 from:</i> • Road surfaces deform as tar melts - need tarmac with a higher melting point • Rail track expands and buckles above the temperature it was pre-stretched for (27 °C in UK) - relay with higher pre-stress • Bridge damage from high river flows / debris; landslides on deforested slopes	2
01.4	• CO₂ is captured from the flue gases (or carbon removed from the fuel before combustion) • Transported by pipeline / ship / tanker and stored in depleted oil or gas fields, aquifers, or used for secondary oil recovery • Limitation: developmental / expensive, uses energy (reducing efficiency), risk of leakage, limited storage sites	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 99-100, 105-108; AQA spec 3.2.1.2.	

Question	Marking guidance	Marks
02.1	• Ozone is very low overnight / early morning, rises from about 08:00, peaks at about 80 units at 14:00 and falls in the evening • Ozone is a secondary pollutant formed by photochemical reactions requiring sunlight (UV), so it peaks in the afternoon after several hours of sunlight • NO₂ (from morning traffic) is broken down by UV to NO and monatomic oxygen; monatomic oxygen reacts with O₂ to form ozone • Ozone lags behind the NO / NO₂ peaks (which are produced by the morning rush-hour traffic) because the reactions take time; falls at night when there is no sunlight	4
02.2	• NO is the primary pollutant produced in engines when nitrogen and oxygen react at high temperature / pressure • NO is oxidised in the atmosphere (by oxygen / ozone) to NO₂, a secondary pollutant	2
02.3	• PANs (peroxyacetyl nitrates) • Eye irritation / breathing difficulty / asthma / emphysema / increased risk of heart attacks (toxic at much lower concentrations than NO_x)	2
02.4	<i>Any 2 from:</i> • Heavy vehicle use giving high NO_x and hydrocarbon emissions • Sunny climate providing the UV for photochemical reactions • Valley / basin topography and frequent temperature inversions trapping pollutants	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 248-249, 253-254; AQA spec 3.4.3.2.3-3.4.3.2.4.	
03.1	• Occupancy sensors detect infrared / movement and switch lights or appliances off when no one is in the room • Programmable thermostats set different temperatures for different rooms and times so unused rooms / periods are not heated	2
03.2	• LEDs convert a much larger proportion of electricity to light rather than heat • Washing machines with cold / low-temperature cycles and fast spins (less drying); dishwashers using less water so less heating; cookers with double / triple glazed doors; fridges with linear compressors; LED screens using half the power of plasma; germanium transistors	3
03.3	<i>Any 3 from:</i> • Turning off lights / appliances rather than standby • Turning thermostats and radiator valves down / off in unused rooms • Only heating the water needed (kettle) • Showering instead of bathing; eco-cycles • Choosing smaller vehicles / less travel	3
03.4	• Stored hot water loses heat continuously through the tank walls and pipes • Heating as needed (or just before) avoids these standing losses; if storage is needed the tank should be well insulated	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 232-233; AQA spec 3.3.4.2.2.	
04.1	• Minimum ozone fell steeply from about 225 DU in 1979 to about 95 DU in the mid 1990s • Since then it has fluctuated around 100-115 DU with a slight recovery / no further decline	2

Question	Marking guidance	Marks
04.2	• $(225 - 95) / 225 \times 100$ • = 57.8% (accept 58%)	2
04.3	• Stratospheric temperatures over Antarctica are the lowest on Earth so stratospheric clouds / ice crystals form • Ice crystal surfaces allow reactions that convert the chlorine released from CFCs into chlorine molecules (Cl_2) during the dark winter • The polar vortex (rotating winds) prevents mixing with the rest of the atmosphere, keeping temperatures low and chlorine concentrated • When sunlight returns in spring (September) it splits the chlorine molecules into free radicals that destroy ozone; levels recover by December as the vortex breaks down	4
04.4	• Ground and satellite UV / ozone measurements show depletion but not the chemistry / correlation is not causation • Air samples collected from the stratosphere by helium balloons and high-flying research aircraft detected chlorine and chlorine monoxide, confirming the Rowland-Molina hypothesis	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 110-111; AQA spec 3.2.1.3.	
05.1	• Its total annual discharge	1
05.2	• Public supplies cannot match consumer demand / the water is often cleaner than rivers • It also reduces urban flooding after heavy rain; surplus can be used for aquifer recharge	2
05.3	• The reservoir is created in the estuary so no farmland / settlements need to be flooded • But it changes important inter-tidal habitats, obstructs shipping and can be polluted by activities anywhere in the river's catchment	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-123; AQA spec 3.2.2.5.	
06.1	• Bacteria in the gut convert nitrates to nitrites • Nitrites react with haemoglobin so it cannot carry oxygen • Infants are most at risk because they have more of the gut bacteria and their haemoglobin reacts faster; usually from shallow wells receiving arable drainage	3
06.2	• Nitrites may be converted in the gut to nitrosamines which are carcinogenic • But most research has been on rats / the evidence in humans is not conclusive; nitrites are also ingested in processed food so the source cannot be isolated	2
06.3	<i>Any 3 from:</i> • Reduce use of nitrate fertilisers / match application to crop need • Avoid ploughing / applying fertiliser at times of heavy rain (autumn / winter) • Use low-solubility / slow-release fertilisers rather than soluble rapid-release ones • Store manure / slurry where it cannot leach or wash into rivers • Grow crops with lower nitrogen requirements (eg triticale instead of wheat) • Grow legumes with nitrogen-fixing bacteria instead of relying on artificial fertiliser	3

Question	Marking guidance	Marks
06.4	- Iron(III) sulfate is added to effluent: phosphates are precipitated as insoluble iron phosphate (a fine sediment that can be used as fertiliser) - Lakes: dredging phosphate-rich sediments / adding iron(III) sulfate to reduce phosphate solubility (eg the Norfolk Broads)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 267-268; AQA spec 3.4.3.2.9.	
07.1	- Ground surface: pitfall trap (container sunk level with the surface, cover, checked after 24 h) - only mobile surface animals; trapped predators eat others; preservative attracts / repels; escapes - Leaf litter: Tullgren funnel (sample on mesh under a light; heat / light drive invertebrates down into a container for a standard time) - only species repelled by heat / light; some die or stop moving; large ones cannot pass the mesh - Branches: beating tray (branch beaten a set number of times over a sheet) - hard to standardise vigour; some species not dislodged / fly away; high branches unreachable	6
07.2	- Moths only fly as adults for part of the year and activity depends on weather / season, so a species may be present but not caught - Brightness / wavelength of the lamp affects attractiveness so catches vary between traps	2
07.3	- The area of ground covered by the suction pipe is known so invertebrates per m² can be calculated - A pooter is only a handling / collection device for individuals already found, not a sampling method	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 397-399; AQA spec 3.7.2.1.	
08.1	• B - Low liposolubility and low persistence	1
08.2	- Bioaccumulation: build-up of a pollutant in the tissues of an individual organism over time because uptake is faster than excretion / breakdown - Biomagnification: increase in concentration of the pollutant at successive trophic levels of a food chain	2
08.3	- They are intended to be toxic (to the target species) - They are deliberately released / spread into the environment (rather than being an accidental or waste release)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	
09.1	- Water = 65.0 - 55.0 = 10.0 g; fresh soil = 40.0 g 10.0 / 40.0 x 100 = 25.0%	2
09.2	- Organic matter = 55.0 - 52.6 = 2.4 g; dry soil = 30.0 g 2.4 / 30.0 x 100 = 8.0%	2
09.3	- So all water is removed and the mass loss on burning is due to organic matter only - Heating until mass no longer changes shows all water has evaporated	2
09.4	- Organic matter releases nutrients as it decomposes and provides food for soil biota - Increases water retention and, as humus, binds particles into crumb peds improving structure / aeration - Surface organic matter / humus adhesion reduce rain splash and wind erosion	3

Question	Marking guidance	Marks
09.5	• Samples taken from different depths / locations (organic matter is concentrated near the surface); incomplete burning; different heating times	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-159, 412; AQA spec 3.2.5.1, 3.7.2.1.	
10.1	• Together they form a dispersed layer in the stratosphere (the ozone layer / ozonosphere) • They absorb ultraviolet light, which drives a dynamic equilibrium of reactions that form and destroy ozone • The UV energy is converted to chemical / heat energy so most UV that passed the upper atmosphere never reaches the surface	3
10.2	<i>Any 2 from:</i> • Provides carbon, oxygen, hydrogen and nitrogen (as N₂, O₂, CO₂, H₂O) for biological molecules • Delays the escape of infrared energy, keeping the surface warm • Distributes heat and water vapour by winds / ocean currents • Provides atmospheric pressure so liquid water can exist • Deflects damaging radiation of the solar wind in the upper atmosphere	2
10.3	• Energy is absorbed at different altitudes (eg UV absorbed in the stratosphere, surface heating warms the troposphere) so temperature gradients change with altitude, producing layers	1

Question	Marking guidance	Marks
10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Troposphere: enhanced greenhouse effect - anthropogenic CO₂ (fossil fuels, deforestation), methane (rice, livestock, landfill), N₂O, CFCs, tropospheric ozone; more infrared absorbed and re-emitted; aerosols and smoke increase albedo / scatter light; land-use change alters surface albedo and evapotranspiration • Stratosphere: CFCs / halons / HCFCs release chlorine and bromine which catalyse ozone destruction, shifting the O / O₂ / O₃ equilibrium; less UV absorbed in the stratosphere, more reaches the surface; stratospheric cooling • Consequences: changes to climate, ocean currents (thermohaline), hydrological cycle, sea level, species distributions; increased UV damage to organisms • Difficulties of prediction: linked processes where one change alters others; feedback (positive and negative) and tipping points; time lags (ocean heat capacity, long residence times of CFCs and CO₂); natural variability (solar output, volcanoes, Milankovitch cycles); incomplete understanding of atmospheric processes; limited historical data so proxies needed; models with different assumptions • Evaluation: some effects well understood (greenhouse physics, ozone chemistry proven, Montreal Protocol success) but scale and timing of regional impacts uncertain; precautionary principle • Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89, 108-113; AQA spec 3.2.1.1, 3.2.1.3.	

Question	Marking guidance	Marks
11.1	Discuss the strategies that can be used to provide sustainable supplies of water for a growing population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Context: uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • New sources: rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • Management: artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • Conservation: low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering; pollution control to protect sources • Treatment: matching quality to use; screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	25

Question	Marking guidance	Marks
11.2	Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Attributes: accuracy, precision, representative samples, reliability, anomalies, validity; hypothesis and null hypothesis; preliminary studies • Sampling design: random vs systematic location and avoiding bias; timing (diurnal, seasonal, long-term, weather); sample size and number and the running mean; standardisation of technique and control of other variables • Techniques for abiotic and biotic factors and their limitations (quadrats, transects, traps, mark-release-recapture assumptions, keys, meters and calibration) • Statistics: variability, standard deviation and error bars, significance and p-values, correlation vs causation; choosing Spearman's rank, chi-squared, t-test, Mann-Whitney U • Application to Paper 1 topics: monitoring pollution (noise, air quality, water), climate proxies, energy surveys, soil analysis • Evaluation of trade-offs (cost, time, disturbance) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.	25